Learning Language & Content by Design

Scaffolding Multilingual Learners' Success in K–12.

Kris Liu

Habeeba Shajeer Shamja

Trinity Western University

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Dr. Carolyn Kristjánsson

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Table of Contents

1. Introduction
2. Literature Review
   2.1 Multilingualism in English–Medium Schooling: Demographic Reality and Structural Responsibility
   2.2 Theoretical Foundations of Second Language Development
   2.3 Identity, Intersectionality, and Educational Equity
   2.4 Academic Language Development and Instructional Frameworks
3. Case Study Context
   3.1 Provincial Context: Multilingual Education in British Columbia
   3.2 Community Context: Linguistic Diversity in Abbotsford
   3.3 School and Classroom Context
   3.4 Learning Needs of Multilingual Learners in the Classroom
4. Lesson Design and Instructional Overview
   4.1 Overview of the Lesson Topic
   4.2 Content Objectives
   4.3 Language Objectives
   4.4 Instructional Framework
   4.5 Supporting Multilingual Learners
5. Justification and Reflection
   5.1 Justification of Instructional Design
   5.2 Individual Reflection — Kris Liu
   5.3 Individual Reflection — Habeeba Shajeer Shamja
6. Conclusion

References

Appendices
   Appendix A — Language Needs Analysis
   Appendix B — Full Lesson Plan
   Appendix C — AI Use Statement
   Appendix D — Collaboration Statement

1. **INTRODUCTION**.

Multilingual learners (MLLs) become an elevated and important presence within English-medium K–12 classrooms across British Columbia and other linguistically different regions of Canada. The consistent and repeated patterns of immigration, mobility as well as transnationalism has taken a moulding on the demographic landscape of schooling, placing the important and mainstream educators at the blending of content instruction and language enhancement. In such scenarios, teachers do have a dual responsibility: making students' mastery of disciplinary knowledge while together aiding and assisting the enhancement of academic English proficiency. Even though many MLLs attain as well as acquire conversational fluency within a comparatively short period of time, the boosting and development of the specialized and abstract language of schooling needs sustained, explicit and contextually responsive instructional assistance. Without deliberate eye to language demands embedded within curricular operations and tasks, multilingual learners may face hurdles to connecting the grade-level content despite having substantial cognitive and experiential resources.

This Applied Research Report understands and evaluates on how language and content learning can be intentionally combined to scaffold multilingual learners' academic success in a Grade 4 science classroom. Taking inspiration upon key theoretical frameworks and pedagogical perspectives talked in EDUC 520, this paper synthesizes research on second language acquisition, identity and equity along with the enhancement of academic language. All these conceptual insights are beautifully applies to a case study of Ten Broeck Elementary School in Abbotsford, British Columbia, with more intentional attention to Grade 4 learners operating at Developing (2) and Expanding (3) proficiency levels as taken in the BC English Language Learner Standards. The report ends up in the design of a three-day scaffolded science lesson on states of matter that strategically blends content objectives with very exclusive and thoughtful language objectives. The lesson is considered in connection to research, contextual considerations along with foundational pedagogical values, preceded by

reflective implications for professional practice and ongoing inquiry.

2. LITERATURE REVIEW.

2.1 Multilingualism in English-Medium Schooling: Demographic Reality and Structural Responsibility.

The active presence of multilingual learners (MLLs) in English-medium K–12 classrooms is not to be considered as an exception but a well defining feature of contemporary schooling. Procedures of globalization, transnational mobility and economic migration have shaped the educational landscapes through Canada, especially in linguistically diverse provinces like British Columbia. Within this emerging and upgraded context, multilingualism is not an anomaly needing special accommodation; more than that, it mirrors wider sociolinguistic realities moulding the modern educational systems. The Douglas Fir Group (2016) conceptualizes second language acquisition (SLA) through a transdisciplinary framework that places language development through micro-level interactions, meso-level institutional scenarios and macro-level sociopolitical encounters. This ecological perspective focuses on the fact that language learning is largely connected within social, cultural, and structural atmospheres.

During such scenarios, mainstream educators make a strong inheriting of a dual and largely complex duty. As Gibbons (2015) claims, teachers are accountable for making possible of the disciplinary knowledge while also aiding the development of academic English. All these are linked and connected. When classroom instruction goes without thorough attention to language demands, it implicitly takes into account that students do have the linguistic resources needed to engage with abstract academic ideas. Such assumptions systematically make a strong privilege upon students whose beforehand schooling and home environments blend with dominant academic registers, while disadvantaged those whose linguistic repertoires fall different from these situations.

The demarcation between conversational and academic language also makes a

clarification over the magnitude of this instructional duty. Cummins (2000) distinguishes between Basic Interpersonal Communicative Skills (BICS) and Cognitive Academic Language Proficiency (CALP), showing that conversational fluency largely boosts within one to two years, at the same time academic proficiency typically needs five to seven years of maintained exposure and structured instructional support. Subsequent research has affirmed this developmental trajectory (Soto-Corominas et al., 2020). Likewise, the repetition of achievement gaps among multilingual learners should not be taken and evaluated as evidence of limited ability, but more than as a predictable result of insufficient scaffolding for academic language enhancement.

Academic language is not only “more difficult vocabulary.” On the other hand, it stands as a specialized register formulated by abstraction, nominalization, discipline-specific terminology, dense information packaging, and syntactic structures (Schleppegrell, 2004, 2007). Mastery of this register makes and necessitates learners to participate completely in the epistemological practices of schooling, comprising explaining, claiming, understanding and bringing the disciplinary knowledge. When these linguistic structures are totally implicit, connection to academic content is stratified along linguistic lines. Due to that, demographic changes in schooling ask for more than incremental pedagogical adjustments; they need a basic reconceptualization of mainstream teaching as language-aware pedagogy.

2.1 THEORETICAL FOUNDATIONS OF SECOND LANGUAGE DEVELOPMENT.

Second language development is largely identified as a complex and socially placed procedure that cannot be minimised to the simple standards of vocabulary and grammar. Contemporary research in second language acquisition (SLA) focuses on the fact that language learning happens through different as well as diversified conversations between cognitive processes, social atmosphere and institutional scenarios. The Douglas Fir Group (2016) states a transdisciplinary framework for SLA that places language learning that goes through three interconnected levels: the micro-level of individual connection and cognition, the meso-level of institutional and community standards and along with that the macro-level of wider sociopolitical systems. This ecological viewpoint focuses on the fact that language development is embedded in the atmospheres where learners make a participation and interact. Coming to the classroom scenarios, sociocultural theory gives a significant theoretical foundation for realising on how language learning happens through mediated interaction. Taking upon Vygotskian principles, sociocultural approaches focus on this side that learning takes place through participation in socially meaningful standards within the Zone of Proximal Development (ZPD). The ZPD refers to the distance between what learners can stand very independently and what they can attain with proper assistance from teachers or peers (Ohta, 2013). Through proper and guided participation, learners slowly internalize linguistic structures and disciplinary methods of thinking that were started modeled and scaffolded during interaction. Language hence operates simultaneously as both the object of learning and the medium through which learning happens. Classroom talks, collaborative activities, and the accelerated participation become central methods through which learners enhance linguistic competence while building the conceptual understanding. Research on classroom interaction also shows that peer dialogue and collaborative learning atmosphere give chances for multilingual learners to make an experiment with evolving language standards, negotiate meaning and get feedback that aids language development (Ohta,

2013).

Another impactful theoretical perspective informing language instruction is Krashen's idea of input. As from this perspective, learners attain language when they are brought to input that is slightly beyond their present standards of proficiency but still realisable within a meaningful scenario (Gibbons, 2015). At the same time, contemporary scholarship indicates that input alone is not sufficient for maintained language development. Learners also need chances to get language, connect with others, and make an active participation in tasks that need them to articulate ideas using largely complex linguistic structures.

Listening and speaking procedures does have an important position in this interactional development. Brown (1994) claims that impactful communication needs learners to make an interpretation of the spoken language, connect appropriately and negotiate meaning through interaction. Instructional practices that motivate on the active participation, collaborative dialogue and scaffolded participation hence bring atmospheres in which language development and content learning happen together. Taken together, these theoretical perspectives focus on the fact that language acquisition is a socially mediated process needing intentional classroom design that aids meaningful interaction and elevated language enhancement.

2.3 IDENTITY, INTERSECTIONALITY, AND EDUCATIONAL EQUITY.

Language learning cannot be realised only as a cognitive procedure; it is largely moulded by social identity, power relations and educational systems. Scholars have largely focused on the side of the idea that multilingual learners' experiences in school are impacted by wider social dynamics that change connection to linguistic and cultural standards. Schleppegrell (2007) claims that when academic language expectations are implicit in classroom instruction, students who do have close connection with dominant discourse practices are made into a close privilege, while others are systematically disadvantaged. This perspective focuses on the side of that language that operates as a gatekeeping methods within educational systems. The concept of intersectionality also complicates very simplistic realisations of multilingual learners. Linguistic identity blends with elements like socioeconomic status, cultural background, and migration history to mould learners' educational experiences. Brisk et al. (2010) figures out that multilingual learners stand as different socioeconomic groups, still structural inequalities often focus linguistic diversity in schools standing as marginalized communities. In such scenarios, students may encounter different overlapping challenges that impact their chances for academic participation and achievement.

Within content-area equipment, these challenges are largely accelerated by the linguistic complexity of disciplinary discourse. Research has indicated that academic subjects like mathematics and science need students to connect the specialized vocabulary, complex sentence structures, and abstract conceptual descriptions (Schleppegrell, 2004). For multilingual learners who are together enhancing the academic English proficiency, these linguistic demands can bring important hurdles to showing their conceptual understanding.

At the same time, recent research focuses on the asset-oriented viewpoints that identify multilingual learners' linguistic and cultural resources as strengths more than deficits. Translanguaging approaches focus on the value of permitting learners to draw upon their complete linguistic repertoires when making interpretations and building

knowledge. Echevarría et al. (2024) claim that strategically blending students' home languages into classroom instruction can assist wide comprehension and make necessary dealings of the connections between before knowledge and new academic ideas. Likewise, Roessingh (2011) shows that literacy practices that bring in cultural narratives and multilingual perspectives mould on both language development and learners' idea of identity.

The affective dimensions of learning has a position in multilingual learners' participation in classroom activities. Fresacher (2016) indicates that positive emotional atmospheres boost the learners' willingness to communicate, take risks, and make an engagement with new language forms. When learners feel psychologically safe and important within the classroom community, they are more inclined to participate in interactions that assist language development. Taken together, these perspectives show that academic language instruction is not only a pedagogical technology but an issue blended to equity and connection. By making disciplinary language seen and explicitly teachable, educators can help assure of the fact that multilingual learners are not excluded from academic participation due to linguistic barriers.

2.4 ACADEMIC LANGUAGE DEVELOPMENT AND INSTRUCTIONAL FRAMEWORKS.

While realising and having an idea on the sociocultural and identity-related dimensions of multilingual learning is needed, impacted classroom practice needs instructional approaches that thoroughly address the linguistic demands blended within academic content. Research consistently shows on the element that multilingual learners does have a benefit from pedagogical frameworks that blend and integrate language development with subject learning more than taking care of language instruction as a distinguishing element of the curriculum (Gibbons, 2015). Within this viewpoint, language is not only an instrument for communication but the primary medium through which disciplinary knowledge is built, divided and kept and expressed.

Systemic Functional Linguistics (SFL) gives a significant analytical framework for realising on the relationship between language and academic information. As by Halliday's functional theory of language, meaning is interpreted and moulded by the connection and interaction between context and linguistic choices, especially through the dimensions of field, tenor, and mode (Schleppegrell, 2004). In school scenarios, these contextual elements give an importance to the specialized registers that make a difference largely from daily conversational language. For example, scientific discourse repeatedly depends on generalized present tense, technical vocabulary, nominalized processes and logical connectors that define on the cause-and-effect relationships. For multilingual learners, attaining these linguistic patterns is needed for being a part meaningfully in disciplinary practices such as defining on the scientific phenomena, deciding the data, and constructing arguments.

Because academic language is largely implicit in classroom instruction, multilingual learners may find it a struggle to access disciplinary knowledge even when they do have strong conceptual knowledge. Gibbons (2015) therefore emphasizes the significance of scaffolding language development within content instruction. Scaffolding stands to temporary instructional ais that makes learners to perform tasks

that would otherwise be far beyond their independent capabilities. As learners attain greater control over academic language, this assistance is very slowly withdrawn, permitting them to participate more freely in disciplinary learning.

One largely identified instructional model that operationalizes these principles is the Teaching–Learning Cycle linked with genre-based pedagogy. This includes a scenario and sequence of stages designed to slowly assist students' realisation and production of academic texts (Derewianka & Jones, 2016). The cycle starts with constructing the field, where teachers make a strong activation prior knowledge and boost on the shared understanding of the topic. During the modelling stage, students understand exemplar texts while teachers explicitly focus on the side of the key linguistic features. In the joint construction phase, teachers and students collaboratively give texts, permitting learners to practice implying newly introduced language structures with guidance. At last, independent construction allows students to give their own texts, showing their developing control of disciplinary language.

In connection to explicit genre instruction, multimodal approaches to learning can also make a strong support to the side of the multilingual learners' engagement with complex academic ideas. Williams et al. (2019) focus on the fact that meaning-making in classrooms happens through multiple modes of communication, consisting of visual, oral, gestural, and written forms. Visual representations like diagrams, graphic organizers, and illustrations can help minimise linguistic hurdles while bringing forth conceptual understanding. These multimodal supports permit on the multilingual learners to access content knowledge even when their standard in academic English is still enhancing.

Classroom interaction has an important place in aiding the language development. Collaborative dialogue gives chances for learners to make a practice of the emerging linguistic forms within supportive social environments. Ohta (2013) shows that peer interaction within the Zone of Proximal Development permits learners to make an experiment with language while getting feedback and support from more impactful speakers. Through such connections, students slowly internalize the linguistic patterns needed for academic participation.

Added together, these instructional frameworks highlight that successful teaching for multilingual learners needs deliberate integration of language and content objectives. By modelling disciplinary language, giving scaffolded opportunities for discussions and operations and making use of the multimodal resources, educators can bring in learning environments in which multilingual learners are able to link academic knowledge while boosting on the proficiency in the language of schooling.

3. CASE STUDY CONTEXT.

3.1 PROVINCIAL CONTEXT: MULTILINGUAL EDUCATION IN BRITISH COLUMBIA.

The linguistic diversity seen as well as analysed in Canadian classrooms is largely mirrored in the province of British Columbia, where multilingual learners become a magnificent and steadily elevating portion of the student population. In recent times, immigration, transnational mobility and global economic migration have made a strong commitment to elevating on the multilingual educational atmospheres across the province. Due to that, schools are no longer formulated by linguistically homogeneous student populations but on the other hand it stands for the diverse linguistic communities in which students stand as the different cultural and linguistic resources to the classroom. All these demographic fluctuations have needed educational systems to make a proper reconsideration of the conventional instructional approaches and to enhance on the policies that assure equitable access to learning for students focusing on the proficiency in English.

The British Columbia Ministry of Education has made a response to these demographic changes by bringing in the English Language Learner (ELL) Standards, which gives educators with a well organised framework for realising and aiding multilingual learners' language development (British Columbia Ministry of Education, 2021). These standards make an outline on the developmental continua across four important spheres of language proficiency: listening, speaking, reading, and writing. More than realising on the side of language learning as distinct from academic learning, the standards identify that multilingual learners boost English proficiency while together engaging with subject-area content.

Likewise, teachers are motivated to blend in language development within daily classroom instruction more than relegating it to distinct language support programs.

The ELL Standards also define phases of language development that focus on teachers in understanding proper instructional support for students at distinct levels of

proficiency. Learners accelerate through developmental stages that begin from evolving to expanding levels of language competence. Students at the Developing (Level 2) stage are mostly able to realise the basic classroom instructions and make a participation in structured interactions with assistance, but they may still depend largely on contextual cues like visuals, gestures, and simplified language. Learners at the Expanding (Level 3) stage show elevating independence in communication and comprehension; at the same time, they may resume to face struggles when connecting on the complex academic texts or giving extended written explanations that need very specialised vocabulary and abstract reasoning.

These developmental phases focus on highlighting the significance of explicit instructional methods that connect both conceptual realisation and linguistic development. Without very focused scaffolding, multilingual learners may struggle to show their information of subject content due to constrained familiarity with the specialized language structures used in academic discourse. Due to that , provincial policy focuses on the duty of all classroom teachers not only language specialists to aid multilingual learners' academic language boosting while teaching curricular scenario.

3.2 COMMUNITY CONTEXT: LINGUISTIC DIVERSITY IN ABBOTSFORD.

The Abbotsford School District, placed in the Fraser Valley region of British Columbia, gives a great instance of the linguistic diversity moulding on many Canadian school circles and communities. Coming over the past several decades, Abbotsford has faced important population elevation and boosting cultural diversity due to immigration and international migration. These demographic trends are mirrored in the district's student population, where many learners are originally coming from multilingual households and use languages other than English in their homes and circles.

All among the languages that are taken within the district, Punjabi is the largely spoken native languages, mirroring the existence of South Asian communities within the place. In connection to Punjabi, students may have a strong speech to the plethora of other languages comprising of the Tagalog, Mandarin, and different Indigenous or heritage languages. All this linguistic diversity boosts on the classroom atmospheres by bringing students to different cultural standards and experiences. At the same time, it stands significant instructional understanding for educators who must make a design on the learning atmosphere that aid both language development and academic standards.

Students within linguistically different communities largely have linguistic repertoires moulded by family, community, and schooling connections. All of them, mostly done learners may have attained prior schooling in English, on the other hand some may be newly arrived in Canada and looking into an unfamiliar educational structure. Even among students who have been several years in English-medium schools, different levels of exposure to academic English may impact how easily they can make an active engagement with classroom tasks needed in the specialized vocabulary or abstract reasoning.

Teachers who are part in such scenarios must hence blend pedagogical approaches that identify linguistic diversity as both a standard and an instructional crisis. More than seeing multilingualism as a deficit to be come over, contemporary educational

viewpoints focus on the significance of accelerating the students' existing linguistic knowledge to assist learning. At the same time, teachers must make sure that students got to get proper guidance in the language practices needed for academic success, mostly in subject areas where disciplinary discourse changes from daily communication.

3.3 SCHOOL AND CLASSROOM CONTEXT.

Ten Broeck Elementary School stands as a student population within the Abbotsford School District. As is widely seen in schools placed within linguistically different communities, classrooms comprise of students who talk on the multiple languages and who take to the distinct cultural backgrounds and educational standards to their learning atmosphere.

Teachers must look into how language proficiency impacts students' active participation in classroom conversations, and mostly with written texts as well as the capacity to show conceptual realisation through academic activities. In classrooms where multilingual learners are there, teachers repeatedly face students who are needed of making an active engagement socially with peers but who may require additional help when understanding academic vocabulary, realisation of the complex instructions, or giving extended written explanations.

The classroom understood in this case study is a Grade 4 class in which a lot of students are boosting proficiency in academic English. Places on the developmental continua taken in the BC ELL Standards, these learners are taken into assumption to be operating within the Developing (Level 2) and Expanding (Level 3) phases of language development. At these standards, students do have evolving confidence in conversational discussions and can participate in classroom connections when given with proper assistance. They may realise teacher explanations that are put forth by visual aids, illustrations, or situational examples. At the same time, they may still encounter crisis when actively engaging with academic tasks that need precise language use or interpretation of unfamiliar terminology.

3.4 LEARNING NEEDS OF MULTILINGUAL LEARNERS IN THE CLASSROOM.

The linguistic challenges encountered by multilingual learners become especially seen within subject spheres that depend largely on very specialised academic language. Science education gives a good example of this procedure. Scientific explanations largely include technical terminology, abstract ideas and complex sentence standards that define relationships like classification, sequence, and cause-and-effect. These linguistic standards often differ largely from the language that is precisely utilised in everyday conversation. For multilingual learners who are boosting on English proficiency and understanding new scientific concepts, these linguistic demands can bring in impactful cognitive challenges. Students must make an interpretation on the unfamiliar vocabulary, realise the conceptual relationships and talk on their realisation using language patterns that may still be encouraging. Without proper instructional assistance, learners may struggle to show their scientific realisation even when they take in the underlying concepts.

Proper instructional approaches hence require the very conscious blending of language development within content instruction. Teachers can aid multilingual learners by taking on the academic language, giving structured chances for collaborative dialogue and blending visual and multimodal resources that make abstract concepts more needed. Graphic organizers, diagrams, illustrations and accelerated discussions can aid learners link new vocabulary with conceptual realisation while minimising the linguistic hurdles linked with complex academic discourse.

In connection to these instructional requirements, the following section familiarises a scaffolded Grade 4 science lesson aimed at states of matter. The lesson is moulded to blend content learning with explicit language instruction, giving multilingual learners with chances to boost both scientific understanding and the academic language needed for becoming a part in disciplinary discourse.

4. LESSON DESIGN AND INSTRUCTIONAL OVERVIEW.

4.1 OVERVIEW OF THE LESSON TOPIC.

In connection to the linguistic and instructional requirements seen in the case study scenario, this Applied Research Report put forth a scaffolded Grade 4 science lesson aimed at the topic of states of matter. The idea of matter and its physical states like solid, liquid, and gas stands as a background scientific concept within the elementary science curriculum. Realising on how matter can persist in distinct forms and fluctuations between these states aids students boost early scientific reasoning about physical properties, understanding and classification.

For multilingual learners, still, the understanding of scientific concepts is largely focused by important linguistic crisis. Scientific explanations depend on specialized vocabulary, abstract ideas and structured forms of reasoning that stand different from daily language use. Terms like solid, liquid, gas, particles, melting, freezing and evaporation familiarise on new conceptual interpretations while needing students to realise how these ideas are anticipated through very specific grammatical and discourse scenarios. On the other hand, impactful science instructions for multilingual learners must ask on both conceptual realisation and academic language development.

The lesson taken in this report hence combined science content with explicit language support. By blending together conceptual learning with scaffolded language instruction, the lesson visions to assure that multilingual learners are capable to connect disciplinary knowledge while boosting the linguistic resources needed for academic participation.

4.2 CONTENT OBJECTIVES.

The primary content objective of the lesson is to aid students enhance a realisation of the three main states of matter and to identify how matter can fluctuate between these states under distinct scenarios. Through focused understanding, discussion, and real scenarios, students will look into the physical properties that differentiate solids, liquids and gases. They understand how temperature changes can impact the changing of matter from one state to another.

• Identify and describe the three primary states of matter: solid, liquid, and gas.

• Recognize observable properties associated with each state of matter.

• Explain simple examples of changes in matter, such as melting, freezing, and evaporation.

• Apply scientific vocabulary to describe observations related to states of matter.

These objectives blend with the Grade 4 science syllabus, which focuses on analysis, classification and definition of natural procedure. The lesson aids both curricular learning goals and the enhancement of background scientific thinking.

4.3. LANGUAGE OBJECTIVES.

In connection to content learning, the lesson addresses language objectives built to aid multilingual learners' boosting of academic English. As talked in the literature review, academic language changes from daily conversational language in its usage of specialized vocabulary, complex sentence systems and discipline-specific discourse scenarios (Schleppegrell, 2004; Gibbons, 2015). For multilingual learners, taking up these linguistic features is needed for participating in classroom learning.

The language objectives for this lesson aim on three key areas: vocabulary development, oral interaction, and explanatory language.

First, students will enhance the familiarity with needed scientific vocabulary connected to states of matter, consisting of the terms like solid, liquid, gas, particles, melting, freezing, and evaporation. Teachers will familiarise these terms through visual assistance, demonstrations and contextualized instances to assure that learners can connect new vocabulary with concrete experiences.

Second, the lesson will give structured chances for oral connections. Through pair discussions and small-group standards, students will practice defining on observations and describing ideas using introduced vocabulary. Sentence frames and guided prompts will aid learners in bringing meaningful responses, minimising on the linguistic hurdles that may limit participation.

Third, students will boost skills in giving simple scientific definitions. Using scaffolded writing activities, learners will make a practice on bringing their ideas using very basic explanatory structures like cause-and-effect statements and comparisons. These language tasks will accelerate students from oral discussion toward written expression, mirroring the progression from spoken to written academic language defined by Gibbons (2015).

4.4 INSTRUCTIONAL FRAMEWORK.

The construction of this lesson takes upon the Teaching–Learning Cycle, a largely identified framework for bringing language development with content patterns. As taught in the literature review, the Teaching–Learning Cycle gives structured standards of instructional phases that assist students' realisation and production of academic language (Derewianka & Jones, 2016). This scaffolded approach is mostly good enough for multilingual learners as it permits teachers to model disciplinary language while slowly changing duty to students.

The lesson sequence combines four key phases of the Teaching–Learning Cycle:
Building the Field:

The lesson starts with activities that make a change to the students' prior knowledge about daily experiences with solids, liquids and gases. Through illustrations and guided discussions, students are familiarised to the foundational ideas of matter while boosting shared vocabulary linked to observable properties.

Modelling the Genre:
During this phase, the teacher familiarises the examples of simple scientific definitions related to states of matter. Students understand how scientific ideas are taken through language, comprising the use of descriptive vocabulary and explanatory connectors like because and when.

Joint Construction:
In the next phase, the teacher and students together construct explanations refining observations linked to states of matter. Through proper questioning and group conversation, students practice using scientific vocabulary and sentence structures to define their ideas.

Independent Construction:

Overall, students imply their enhancing knowledge by giving short explanations defining the properties of states of matter. At phase stage, scaffolding is slowly minimised, permitting learners to show their growing grip of both conceptual understanding and academic language.

4.5 SUPPORTING MULTILINGUAL LEARNERS.

The lesson design blends a lot of strategies mostly focused to aid multilingual learners' engagement with both content and language. Visual aids like diagrams and demonstrations will aid students link abstract concepts with concrete experiences. Graphic organizers will help learners in bringing their ideas and identifying relationships between different states of matter. The structured classroom dialogue will motivate collaborative learning and give chances for multilingual learners to make a practice evolving language skills in aiding environments. Peer interaction allows learners to connect meaning, clarify misunderstandings and boost trust in using academic vocabulary.

By bringing on these methods with explicit modelling of scientific language, the lesson visions to bring an inclusive learning atmosphere in which multilingual learners can connect scientific concepts while developing proficiency in academic English.

5. JUSTIFICATION AND REFLECTION.

5.1 JUSTIFICATION OF INSTRUCTIONAL DESIGN.

The design of the Grade 4 science lesson on states of matter is focused by research on second language acquisition, academic language enhancement and instructional standards for multilingual learners. As focused in the literature review, multilingual learners make strong benefits from pedagogical standards that bring language learning with disciplinary content more than understanding language instruction as an isolated element of the syllabus (Gibbons, 2015). Likewise, the lesson design thoughtfully blends conceptual science learning with explicit language scaffolding in connection to assisting multilingual learners' activation with both scientific information and academic English.

An important principle accelerating the instructional design is the identification that academic language is different largely from daily conversational language. Scientific discourse repeatedly includes vocabulary, abstract ideas and structured standards of description that need learners to bring an organisation in information using very clear linguistic patterns (Schleppegrell, 2004). For students who are boosting proficiency in English, these linguistic demands may bring in hurdles to showing conceptual understanding. Due to that, the lesson blends targeted vocabulary instruction and maintained chances for language use that allow learners to slowly enhance connection with scientific terminology and explanatory language.

The usage of the Teaching–Learning Cycle gives a framework for scaffolding students' engagement with disciplinary language. This pedagogical approach permits teachers to move from shared experiences toward independent language production (Derewianka & Jones, 2016). During the construction of the field stage, students are familiarised to the topic through illustrations, visual representations, and guided conversations that make an activation before there is information about solids, liquids and gases. This stage permits multilingual learners to link new scientific concepts with experiences while boosting a shared vocabulary for talking on these ideas.

The modelling stage aids learners by illustrating on how scientific explanations are constructed. Teachers aim on the linguistic features utilised in scientific discourse comprising of the descriptive vocabulary, classification language, and causal connectors like because, when, and therefore. By taking these language patterns seen, teachers aid multilingual learners identify how scientific ideas are brought into an organisation and communicated. In the joint construction phase, students work together with the teacher and peers to give explanations about the properties of distinct states of matter. This stage brings chances for guided practice in which learners can make an experiment with very unique vocabulary and sentence structures within aiding learning atmosphere. Collaborative dialogue makes students to negotiate meaning, open upon misunderstandings, and attain very quick feedback on their language use. Lastly , the independent construction stage permits students to imply their boosting knowledge by giving short explanations describing scientific understandings. At this phase, scaffolding is minimised so that learners can show elevating independence in both conceptual realisation and language utilisation. This progression mirrors the gradual release of responsibility model, which aids learners' movement from guided participation toward autonomous learning.

In line to scaffolded instruction, the lesson brings together multimodal learning strategies that make scientific concepts more connected to multilingual learners. Visual supports like diagrams, demonstrations and graphic organizers permit students to link abstract scientific ideas with representations. Research indicates that multimodal approaches boost by making learners to actively engage with information through multiple channels of meaning-making (Williams et al., 2019). For multilingual learners who may still be boosting in the learning of the academic English proficiency, these give important assistance for realising on complex concepts. Collaborative interaction has an important position in aiding language development. Sociocultural perspectives on learning that language acquisition happens through meaningful participation in social blending (Ohta, 2013). Pair and group discussions within the lesson also give opportunities for multilingual learners to practice emerging language skills while gaining promise from peers and teachers. Through such

conversations, students are good enough to rehearse disciplinary language and internalize the linguistic structures required for academic communication.

Adding to that, the lesson design blends with the BC English Language Learner Standards, which focuses on the importance of bringing together language development within subject-area learning. By asking on both conceptual and linguistic ideas of scientific understanding, the lesson aids multilingual learners at the Developing and Expanding levels of language proficiency. This assures that instructional strategies respond to the language requires identified within the classroom scenario.

5.2. INDIVIDUAL REFLECTION – Kris Liu.

Reflecting on the design of this lesson has brought the efforts to understand on the importance of identifying the inseparable relationship between language and content acquisition. In linguistically different classrooms, teachers cannot take an assumption that students will automatically attain academic language through exposure alone. On the other hand, educators must thoughtfully design learning experiences that give structured chances for students to make an active engagement with disciplinary language while boosting the conceptual understanding.

The process of enhancing this lesson has also focused on the side of the importance of scaffolding as an important pedagogical strategy. By slowly aiding students' movement from shared experiences toward independent expression, teachers can bring learning atmospheres that minimise linguistic hurdles and motivate active participation. This approach acknowledges that multilingual learners do have important cognitive and cultural resources that can aid classroom learning when supported properly.

From a wider professional view, this project has focused on the responsibility of educators to bring inclusive learning atmospheres that identify linguistic diversity as an asset more than an inhibition. Effective teaching for multilingual learners needs precise attention to how language operates within academic activities and how instructional methods can make these language demands more linked. Moving forward, this understanding will influence future instructional planning by encouraging greater attention to language objectives alongside content goals.

5.3 INDIVIDUAL REFLECTION– Habeeba Shajeer Shamja

Engaging in the process of designing this lesson has provided valuable insight into the complex relationship between language development and subject learning. One key realization emerging from this project is that multilingual learners often possess strong conceptual understanding but may struggle to demonstrate their knowledge due to the linguistic demands of academic discourse. Recognizing this distinction is essential for educators seeking to create equitable learning environments.

The lesson design process has also emphasized the importance of adopting instructional strategies that actively support students' participation in classroom dialogue. Chances for collaborative discussion permit multilingual learners to practice evolving language skills while boosting confidence in talking on their ideas. Such connections not only aid language development but also bring in a thought of adding on within the classroom community.

In line with this, this project has brought in the significance of seeing multilingualism as a valuable resource within educational scenarios. Students' linguistic and cultural backgrounds give rich chances for meaningful learning experiences when blended into classroom instruction. By taking up inclusive pedagogical approaches that make an acknowledgement of the linguistic diversity, educators can support both academic development and students' idea within the learning environment.

6. **CONCLUSION**.

SHARED CONCLUSION.

This Applied Research Report has understood the position of language-aware pedagogy in aiding the multilingual learners' victory within English-medium classrooms. As linguistic diversity resumes to mould contemporary educational scenarios, teachers are needed to ask on the dual duty of making a facilitation both disciplinary learning and academic language enhancement. The literature reviewed in this report lights upon the fact that multilingual learners' educational experiences are moulded by a complex connection of sociocultural, linguistic and institutional elements. Research on second language acquisition focuses on the side that language learning is not only an individual cognitive procedure but a socially regulated procedure that happens through connections, scaffolding and participation in thoughtful learning standards (Douglas Fir Group, 2016; Ohta, 2013).

The literature also aids the significance of explicitly understanding the linguistic demands blended within academic subjects. Scientific discourse just like many disciplinary intentions of communication depends on specialized vocabulary, abstract reasoning and well organised explanatory language that changes from daily conversational blending (Schleppegrell, 2004). When these linguistic expectations are implicit, multilingual learners may face hurdles to connecting academic knowledge even after having conceptual realisation. As a result, impeccable instruction needs teachers to make disciplinary language visible and connected through thoughtful scaffolding and explicit modelling (Gibbons, 2015).

The case study context taken in this report shows on how these theoretical ideas apply within a Grade 4 classroom in a linguistically distinct school community. Within the Abbotsford School District, multilingual learners stand as a portion of the student population, mirroring on the wider demographic patterns across British Columbia. Students at the Developing and Expanding phases of language proficiency needs instructional connections that assist conceptual learning and language development.

On learning these, the lesson design taken in this report blends science content with explicit language scaffolding to assure that multilingual learners can make a participation meaningfully in classroom learning. The stated science lesson on states of matter shows on how research-informed pedagogical frameworks can be implied in classroom practice. By taking upon the Teaching–Learning Cycle, the lesson gives chances for students to construct background information, understand instances of disciplinary language, collaboratively construct explanations and slowly enhance independence in giving academic language. Multimodal assistance and collaborative dialogue also boost accessibility by making learners to activate with scientific ideas through multiple standards of communication.

Blended together, the literature, case study analysis, and instructional design taken in this report focuses on the significance of seeing language development as an important element of disciplinary learning. When educators blend language and content objectives, they bring in learning atmospheres that permit multilingual learners to connect complex academic concepts when enhancing the linguistic resources needed for academic participation.

Individual Conclusion – Kris Liu.

Reflecting on this research and lesson design procedure has widened my realisation of the role that language has in moulding students' connection to academic learning. One of the most significant insights attained through this project is the identity that impactful teaching for multilingual learners needs careful attention to the language demands taken within classroom activities. Teachers must accelerate beyond the assumption that language development will happen automatically and on the other hand bring in intentional opportunities for students to engage with disciplinary language.

The process of blending language objectives into content ideas has also focused on the significance of scaffolding as an important element of inclusive pedagogy. By

slowly aiding students' movement from participation to independent act, teachers can bring in classrooms where multilingual learners encounter confident taking part in academic discussions and activities. In future teaching standards, I will continue boosting on the instructional strategies that explicitly aid language development while bringing collaborative and inclusive learning atmospheres.

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Appendix A

Language Needs Analysis for Grade 4 Science Lesson: States of Matter.

Language Feature	**Example from Lesson**	**Why it may be challenging for multilingual learners**
Technical vocabulary	solid, liquid, gas, particles,evaporation,melting ,freezing	These terms represent scientific concepts that may not exist in learners ' everyday vocabulary. Students may need visual demonstrations and contextual explanations to understand the meanings of these words
Academic verbs	Describe, observe, explain, compare, identify	These verbs are commonly used in academic instructions and assessment tasks. Multilingual learners may understand the concept but struggle to interpret what the task requires linguistically.
Nominalization (abstract nouns)	evaporation, condensation, transformation	These words convert processes into abstract nouns, which are typical

		of scientific discourse but may be unfamiliar to learners who are still developing academic language proficiency.
Logical connectors	Because therefore, when, if	Scientific explanations often rely on cause-and-effect relationships. Learners must understand how connectors organize reasoning within sentences.
Complex sentence structures	When heat is added, a solid melts and becomes a liquid.	These sentences include dependent clauses that describe relationships between ideas. Multilingual learners may find these structures difficult to interpret or produce independently.
Descriptive scientific language	A solid has a fixed shape, while a liquid takes the shape of it's container.	Comparative language and descriptive phrases require students to organize information using precise vocabulary and grammar.

Explanatory language	Water changes into Vapor when it is heated	Scientific explanations require students to describe processes and relationships between ideas using formal language.
Classroom interaction language	Turn to your partner and explain what happens when ice melts.	Learners must understand instructional language used in collaborative activities and participate in academic discussions using emerging language skills.

The language analysis taken above understands key linguistic features taken within the Grade 4 science lesson on states of matter. Scientific discourse needs students to actively participate with specialized vocabulary, abstract terminology, and structured explanatory language. For multilingual learners, these linguistic demands can bring in challenges when taking into interpretation of the instructions, realising on the scientific concepts, or giving out their ideas using academic language. Due to that, the lesson design blends through instructional supports such as visual standards, illustrations, sentence frames along with collaborative conversations to help learners enhance both conceptual understanding and the language needed for scientific explanations.

Appendix B.

Full Lesson Plan

Lesson Title

Understanding States of Matter

Grade Level

Grade 4

Subject

Science

Topic

States of Matter (Solids, Liquids, and Gases)

Duration

Three 45–60 minute class periods

Curriculum Connection

This lesson aligns with the British Columbia Grade 4 Science curriculum, which focuses on developing students' understanding of matter and its properties. Students explore how materials can exist in different states and how these states can change through processes such as melting, freezing, and evaporation.

Content Objectives

By the end of the lesson sequence, students will be able to:

• Identify and describe the three main states of matter: solid, liquid, and gas.

• Recognize observable properties of each state of matter.

• Explain simple examples of changes in matter, such as melting and evaporation.

• Use scientific vocabulary to describe observations about matter.

Language Objectives

Students will be able to:

• Use key scientific vocabulary such as solid, liquid, gas, particles, melting, and evaporation.

• Describe observations using simple explanatory sentences.

• Participate in pair or group discussions to explain scientific ideas.

• Write a short explanation describing the properties of different states of matter.

Materials

• Ice cubes

• Clear containers of water

• Balloons (to demonstrate gas)

• Chart paper or whiteboard

• Visual diagrams of solids, liquids, and gases

• Graphic organizers

• Vocabulary cards with images

• Science notebooks

Instructional Framework

Teaching–Learning Cycle

Day 1: Building the Field

Introduction (10 minutes)

The teacher begins the lesson by asking students about everyday experiences with matter.

Example questions:

• What happens when ice is left outside the freezer?

• What happens when water is heated?

• What is air?

Students share their ideas while the teacher records responses on the board.

Demonstration Activity (20 minutes)

The teacher demonstrates:

• An ice cube (solid)

Water in a container (liquid)

• Air inside a balloon (gas)

Students observe each example and discuss differences between them.

Key questions:

• Does the object keep its shape?

• Does it flow?

• Can we see it?

The teacher introduces vocabulary:

solid – keeps its shape

liquid – takes the shape of its container

gas – spreads to fill space

Visual diagrams are used to support understanding.

Water in a container (liquid)

• Air inside a balloon (gas)

Students observe each example and discuss differences between them.

Key questions:

• Does the object keep its shape?

• Does it flow?

• Can we see it?

The teacher introduces vocabulary:

solid – keeps its shape

liquid – takes the shape of its container

gas – spreads to fill space

Visual diagrams are used to support understanding.

Guided Discussion (15 minutes)

Students work with a partner to describe the differences between solids, liquids, and gases using sentence frames.

Examples:

• A solid is _____.

• A liquid changes shape when _____.

• A gas spreads because _____.

Day 2: Modelling and Joint Construction

Review Activity (10 minutes)

The teacher reviews vocabulary from the previous lesson.

Students match vocabulary cards with pictures representing solids, liquids, and gases.

Teacher Modelling (15 minutes)

The teacher models how to explain a scientific process.

Example explanation:

“When ice is heated, it melts and becomes a liquid.”

The teacher highlights the language used to explain cause and effect.

Words such as when, because, and changes into are introduced.

Collaborative Activity (20 minutes)

Students work in small groups to observe examples of matter.

Each group discusses:

• What state of matter they see

• What properties it has

• How it might change

Students complete a graphic organizer describing each state of matter.

Day 3: Independent Construction

Writing Activity (20 minutes)

Students independently write a short explanation describing the states of matter.

Prompt example:

“Describe the properties of solids, liquids, and gases. Give one example of each.”

Students are encouraged to use scientific vocabulary introduced during the lesson.

Sharing and Discussion (15 minutes)

Students share their explanations with a partner or small group.

Peers provide feedback and discuss their observations.

Assessment

Assessment will include:

• Observation of student participation during discussions

• Completion of graphic organizers

• Written explanation describing the states of matter

The teacher evaluates whether students:

• understand the concept of states of matter

• correctly use scientific vocabulary

• demonstrate basic explanatory language.

Differentiation for Multilingual Learners

Several strategies are used to support multilingual learners:

• Visual diagrams and demonstrations

• Vocabulary cards with images

• Sentence frames to support oral explanations

• Collaborative learning through pair and group discussions

• Graphic organizers to help structure ideas

These supports help learners connect scientific concepts with academic language while reducing linguistic barriers.

Appendix C.

AI Use Statement.

AI Use Level: 2.

Artificial intelligence (AI) tools were made in the preparation of this Applied Research Report to support editing, organization, and language clarity. AI assistance helped refine sentence structure, improve academic tone, and organize sections of the report according to the required structure of the assignment.

All conceptual ideas, interpretations of course readings, analysis of the case study, and instructional design decisions were developed by the authors. The authors reviewed, revised, and verified all content to ensure accuracy, originality, and alignment with the course materials. The use of AI tools did not replace the authors' independent thinking or engagement with the assigned readings and theoretical frameworks.

The authors take full responsibility for the content, analysis, and conclusions presented in this report.

Appendix D.

Collaboration Statement.

This Applied Research Report was done collaboratively by Kris Liu and Habeeba Shajeer Shamja as part of the needs for EDUC 520 A: K–12 English Teaching Methods. Both authors made an active participation in the planning, research, writing, and revision of the report to ensure a coherent and unified final document.

The authors worked together to review the assigned course readings, analyze the case study context, and design the scaffolded Grade 4 science lesson. The literature review, case study analysis, and lesson design sections were developed collaboratively, with both authors contributing ideas, reviewing sources, and refining the academic arguments presented in the report.

Each author also completed individual reflective components included within the paper. These sections allowed each contributor to reflect on their professional learning and pedagogical perspectives while maintaining the shared focus of the overall report.

Throughout the writing process, both authors communicated regularly to review drafts, provide feedback, and ensure that the final report maintained a consistent academic voice and structure. The authors jointly reviewed the entire document prior to submission to confirm that all sections accurately represented their collaborative work.

Both authors confirm that the work presented in this report represents an equitable and collaborative effort and that all sources have been appropriately acknowledged.